

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/FINAL/C_DEB_AWYe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3333, C2 = 6669, C3 = 1210)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,      0.2, 0.0001,      0.8);
Distrib (pAm,         TruncNormal_cv,        100,      0.2,      40,    2000);
Distrib (r_pAm_pM,    TruncNormal_cv,        0.8,      0.1,      0.1,      2);
Distrib (v,           TruncNormal_cv,        0.18505,   0.1,      0.001,      2);
Distrib (kapJ,        Uniform,                0,         1);
Distrib (kapA,        Uniform,                0,         1);
Distrib (Ehp,         TruncNormal_cv,        100,      0.2,      20,    300);
Distrib (E_coc,       TruncNormal_cv,        30,      0.2,      1,    300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001,   0.3);
Distrib (dv,          TruncNormal_cv,        2,       0.5,      0,    10);
Distrib (de,          LogUniform,          0.000000001, 0.1);
Distrib (tau,         TruncNormal_cv,        7,       0.5,      0,    50);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter, seeds)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

Fm.1. pAm.1. r_pAm_pM.1. v.1. kapJ.1. kapA.1.

Result_mode	0.16278900	273.65900	0.24622400	0.20397300	0.94055900	0.40448900
2.5%	0.09588270	252.21675	0.13291655	0.16761475	0.87740920	0.23095300
97.5%	0.27513390	299.58000	0.28726730	0.23468300	0.95229500	0.46641600
Min	0.00130340	68.84000	0.10027600	0.13535300	0.36368900	0.09830740
Max	0.29993000	313.82600	0.86386800	0.27800300	0.96126100	0.62268800
Result_mean	0.16219424	274.65515	0.21192399	0.20069639	0.92016471	0.35358685
Result_sd	0.04831259	12.51543	0.04099861	0.01723749	0.02117705	0.06135022
	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	dv.1.	de.1.	tau.1.
Result_mode	30.11860	13.18160	0.0048452400	7.4728700	1.271560e-09	19.657900
2.5%	24.39460	11.36470	0.0032738600	5.9806675	1.146070e-09	15.136880
97.5%	66.98820	18.64890	0.0088991650	8.6471800	7.130544e-07	26.273660
Min	20.12680	9.87515	0.0000282676	0.6500460	1.007700e-09	3.506140
Max	93.58210	35.97540	0.0127330000	9.9875600	1.453770e-05	31.133800
Result_mean	40.65953	14.80527	0.0056835259	7.2199053	1.004186e-07	20.869472
Result_sd	11.27818	1.89921	0.0014884698	0.6682898	2.491635e-07	2.819641
	Sigma_W.1.					
Result_mode	0.15264100					
2.5%	0.13640900					
97.5%	0.17585940					
Min	0.12030000					
Max	6.12797000					
Result_mean	0.15511026					
Result_sd	0.05543462					

Number of observations, n = 277
 Number of parameters, k = 12
 Log-likelihood, LL = -64.23663
 BIC = 195.9615

Potential scale reduction factors:

	Point est.	Upper C.I.
de.1.	1.27	2.05
dv.1.	1.02	1.03
E_coc.1.	1.10	1.30
Ehp.1.	1.30	1.84
Fm.1.	1.02	1.07
kapA.1.	1.19	1.54
kapJ.1.	1.31	1.85
pAm.1.	1.04	1.12
r_pAm_pM.1.	1.18	1.51

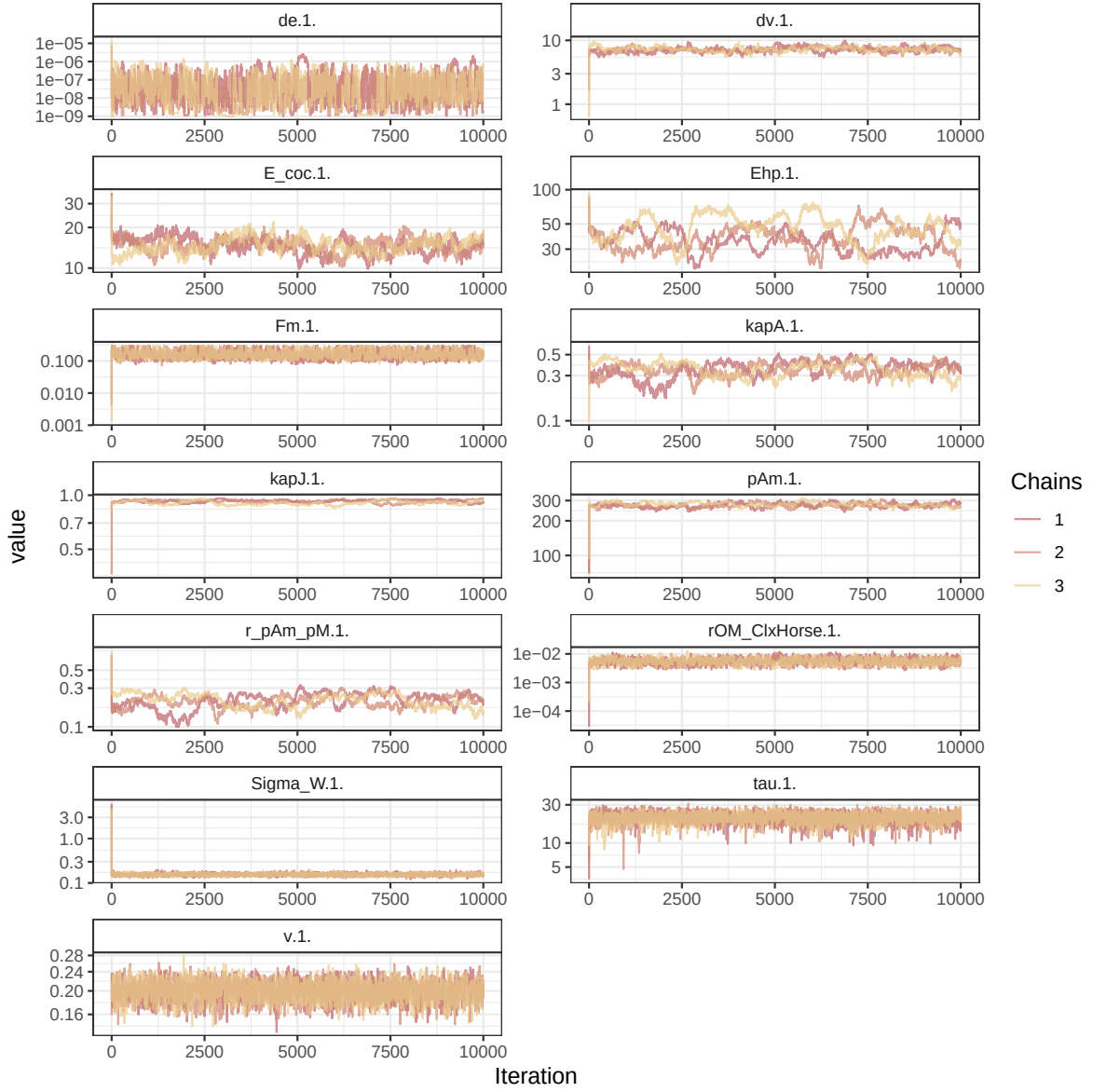


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

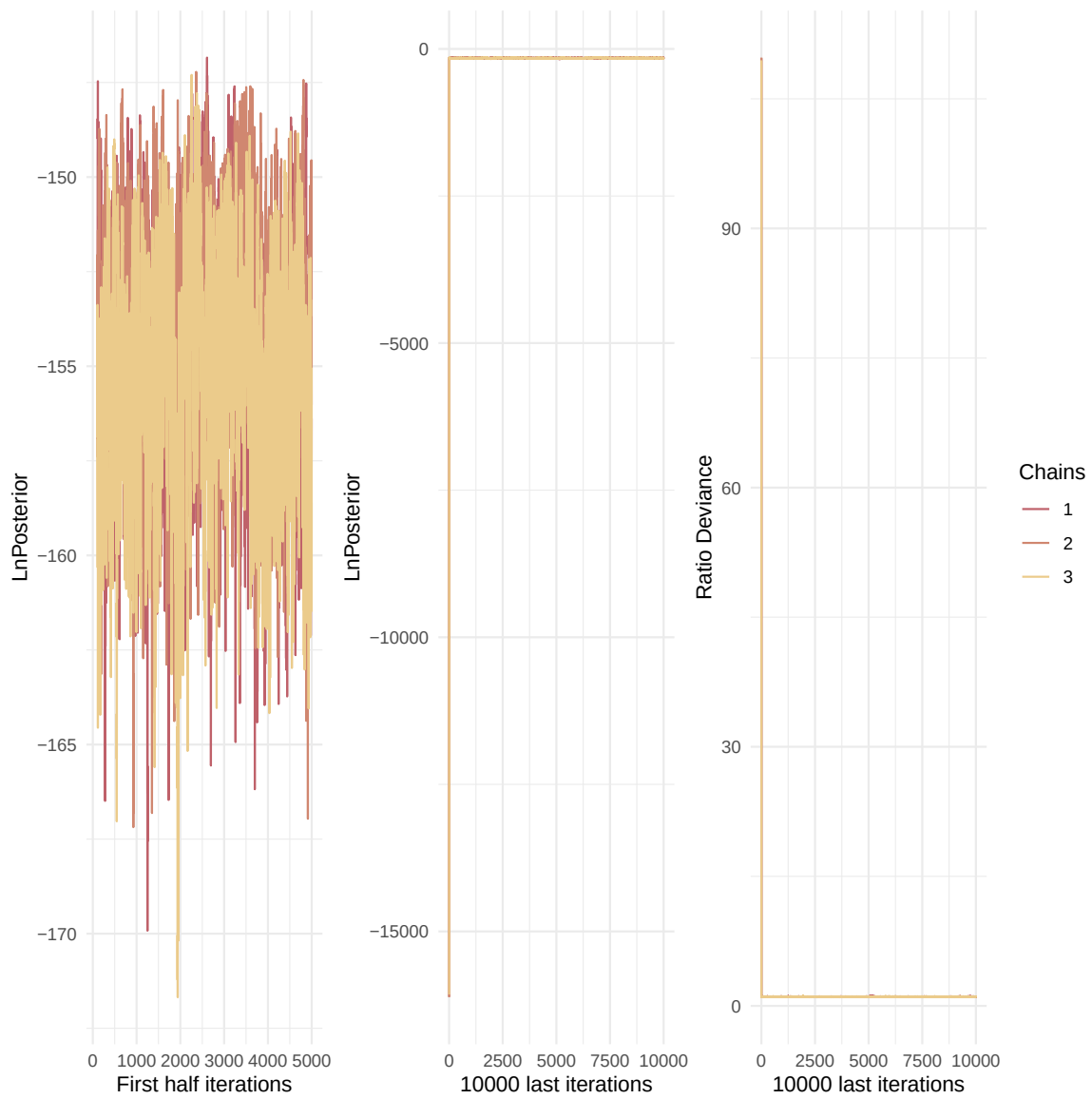


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

rOM_ClxHorse.1.	1.06	1.19
Sigma_W.1.	1.00	1.01
tau.1.	1.01	1.03
v.1.	1.00	1.01

Multivariate psrf

1.31

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

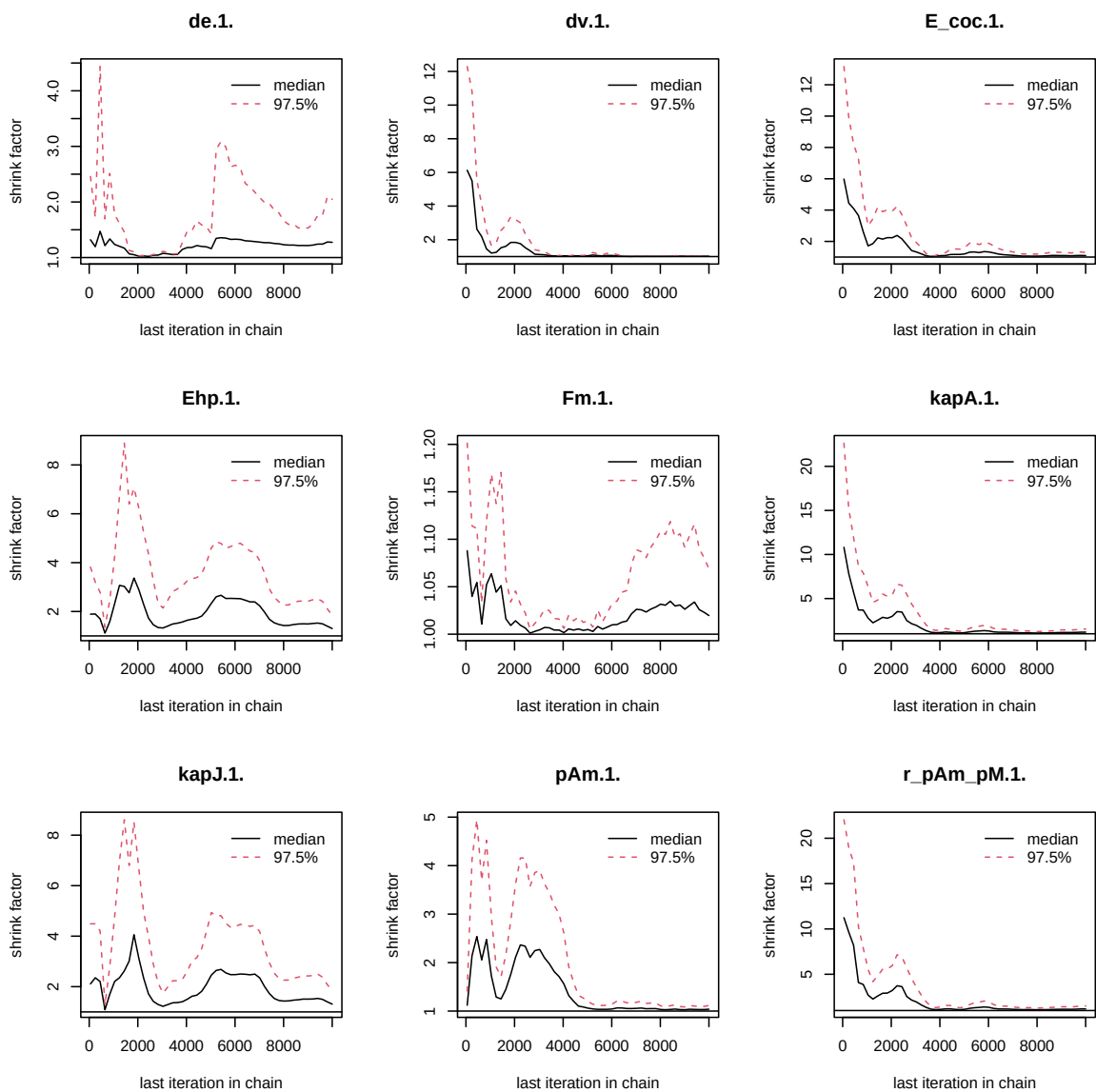


Figure 3: Chains convergence.

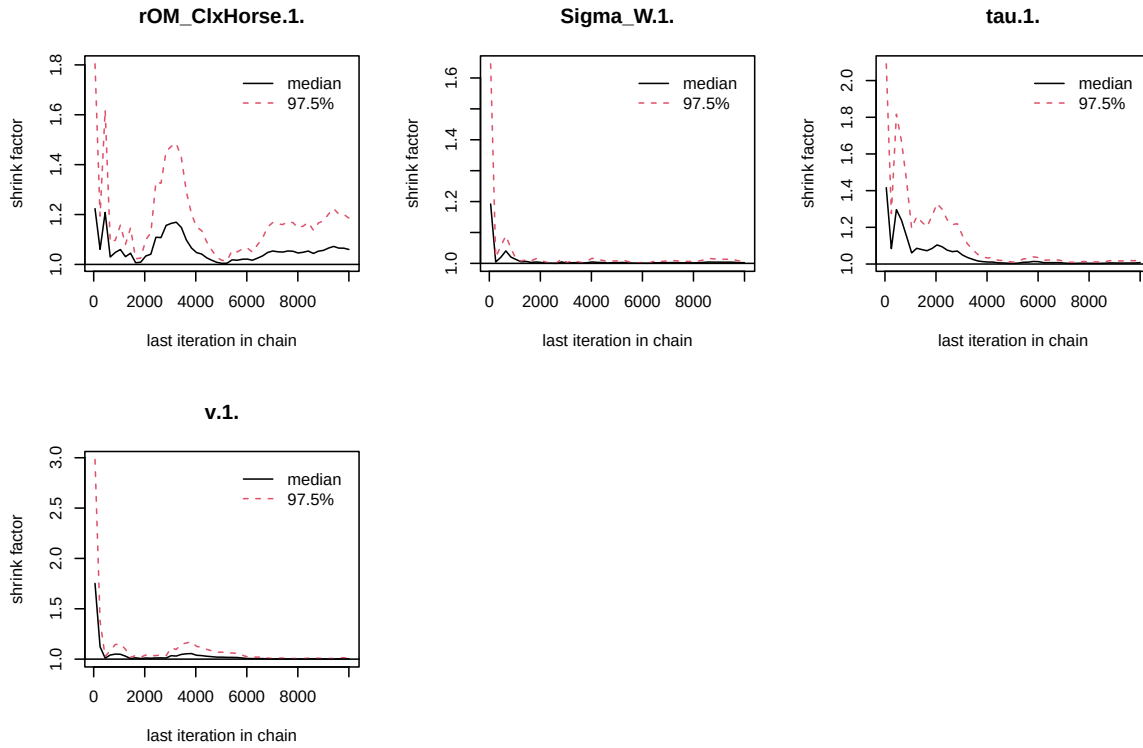


Figure 4: Chains convergence.

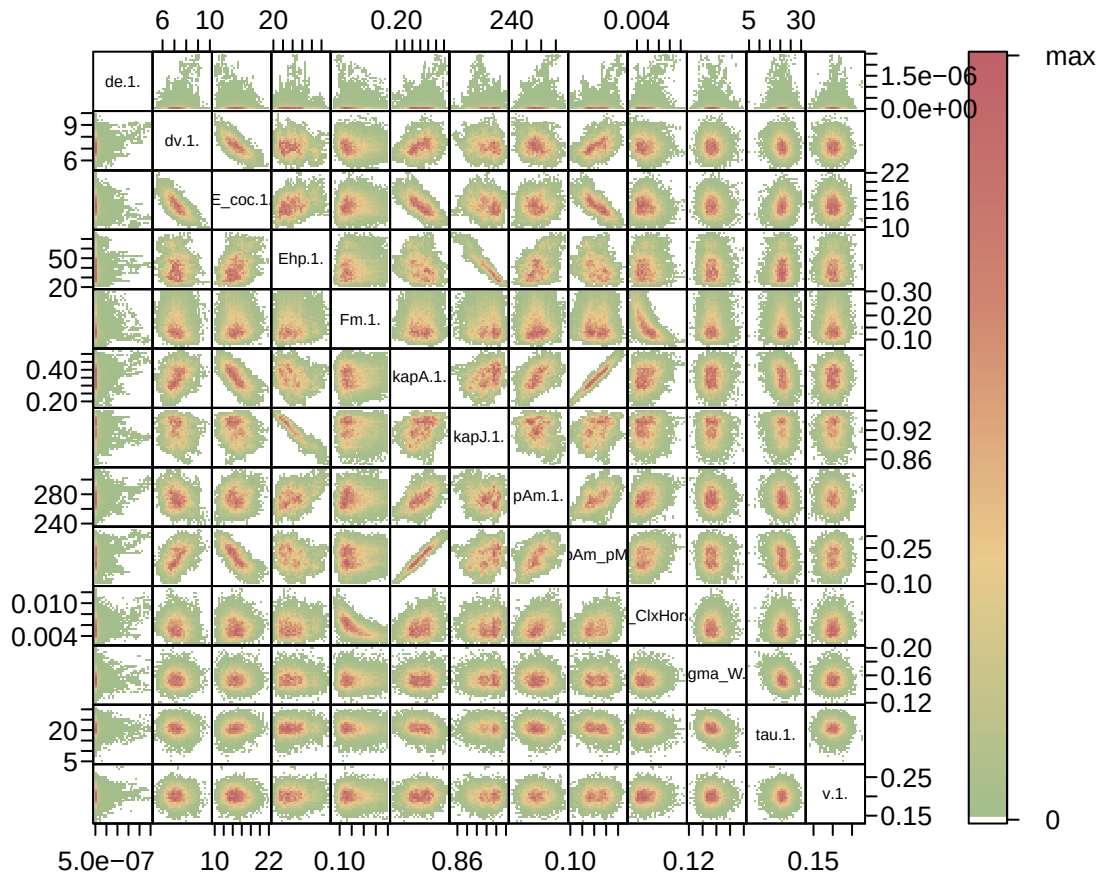


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

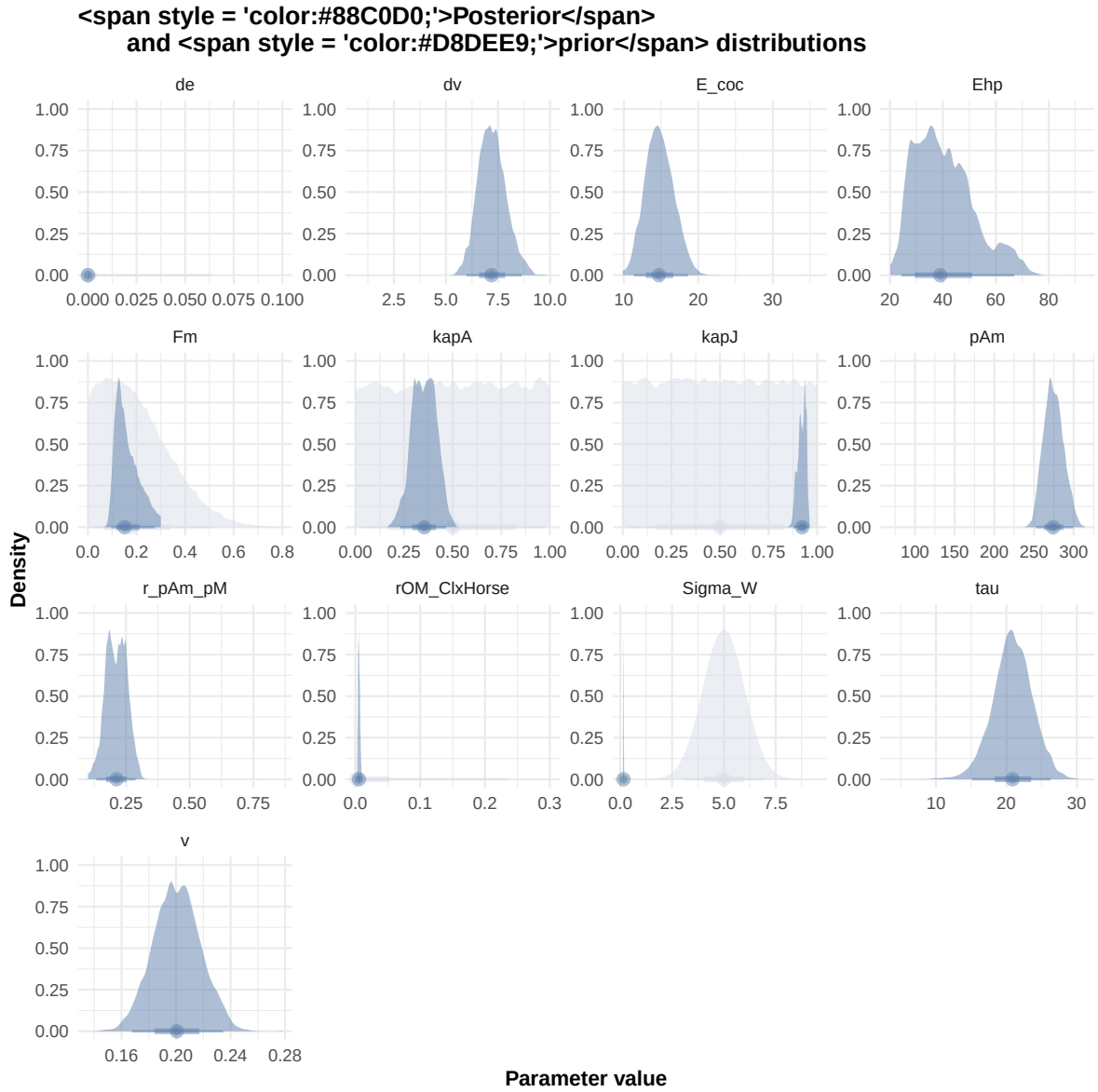


Figure 6: Distributions of the priors and the posteriors.


```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

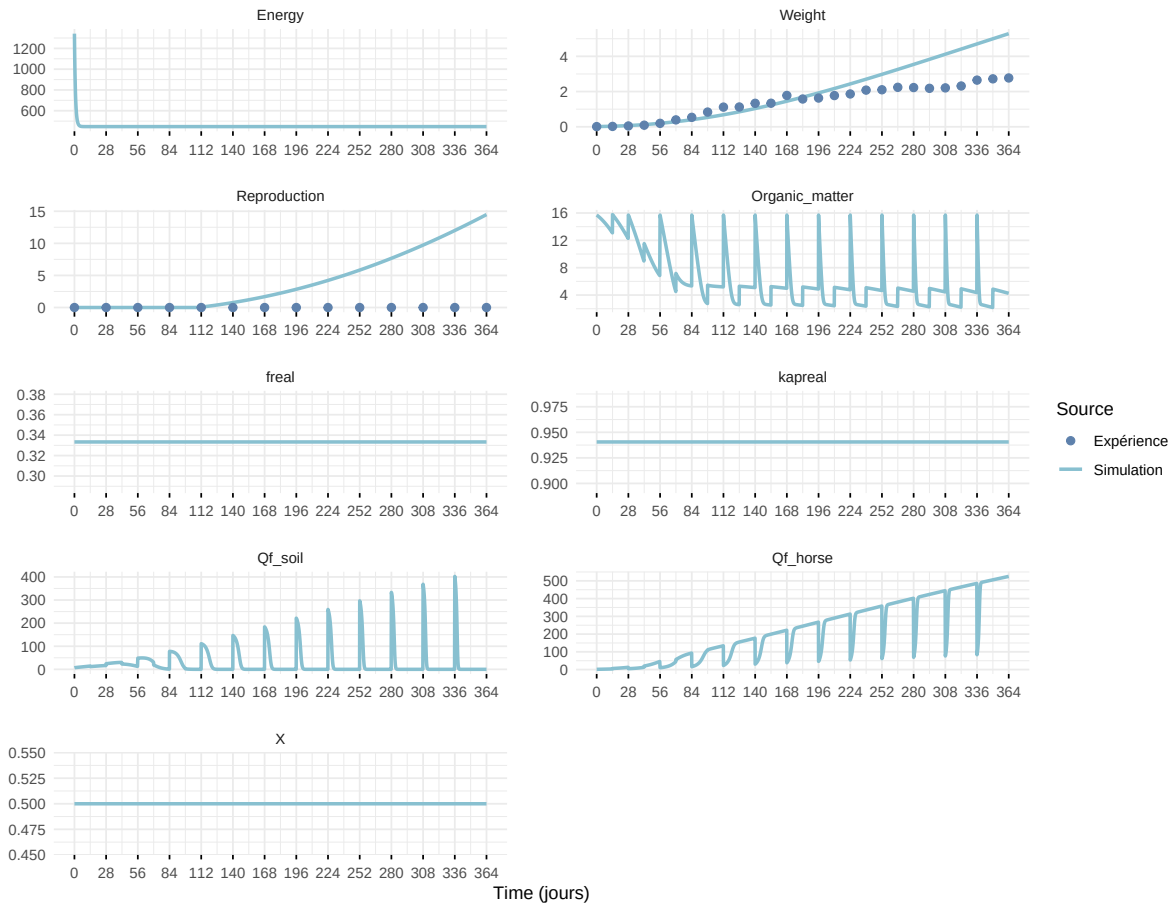
df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected join
 i Row 1 of `x` matches multiple rows in `y`.
 i Row 1 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship =`

"many-to-many" to silence this warning.

Expérience : Bart2019_XP1_1



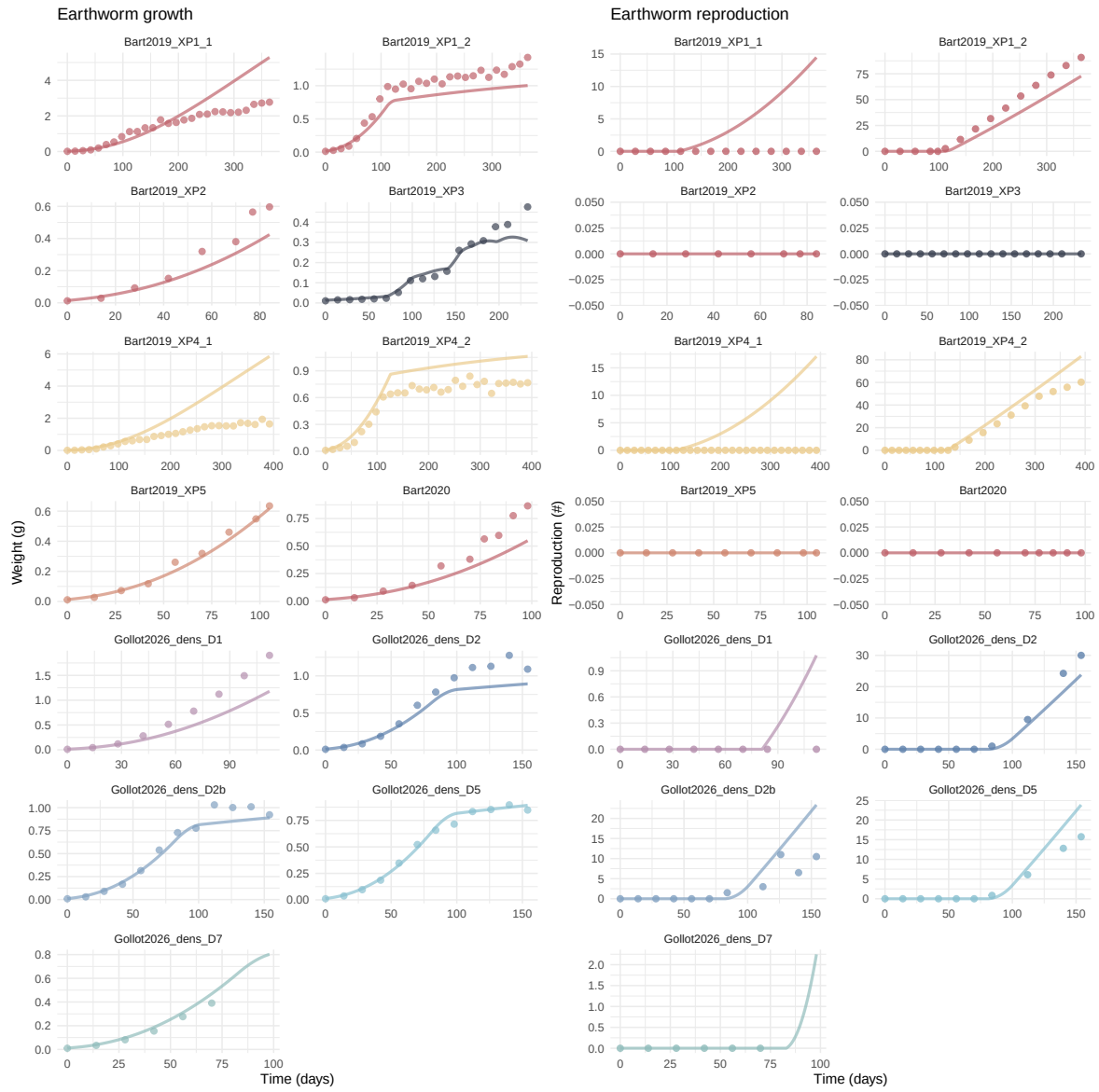
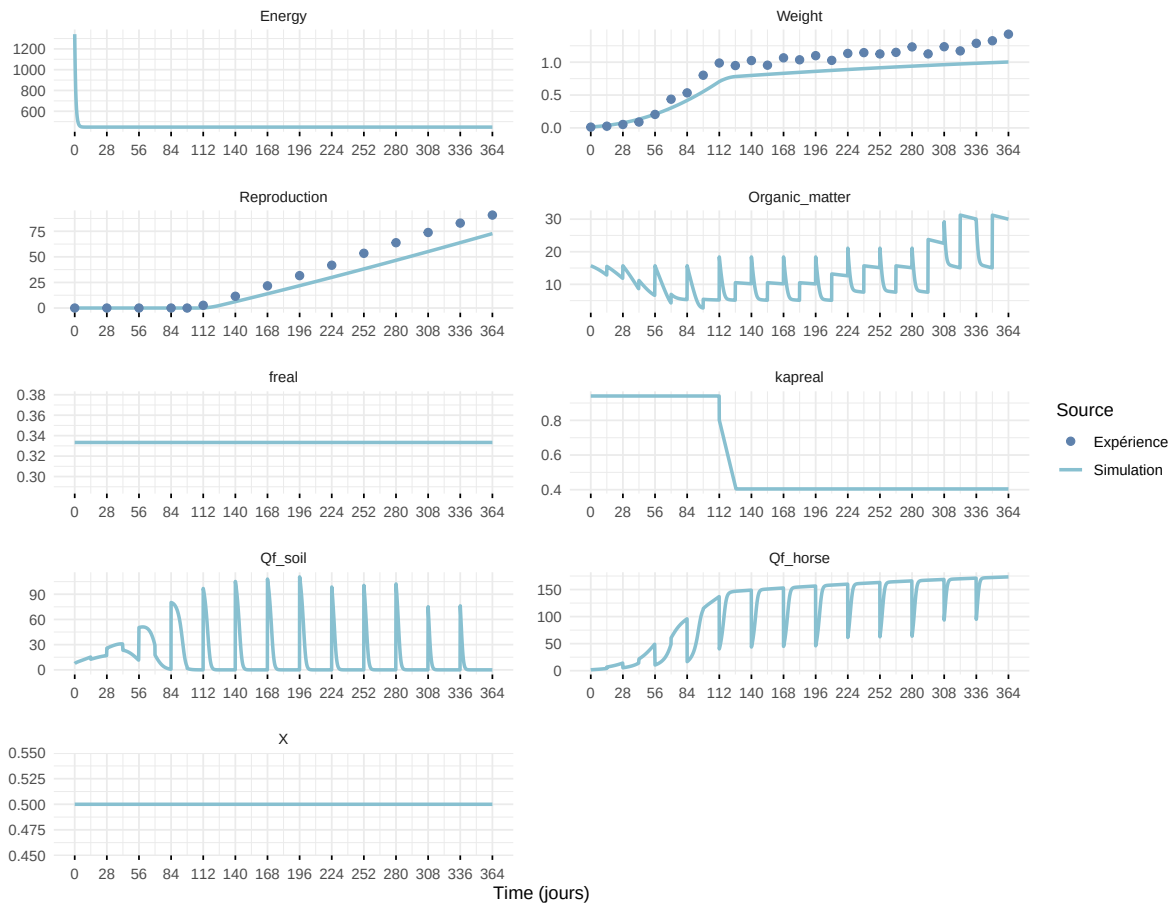
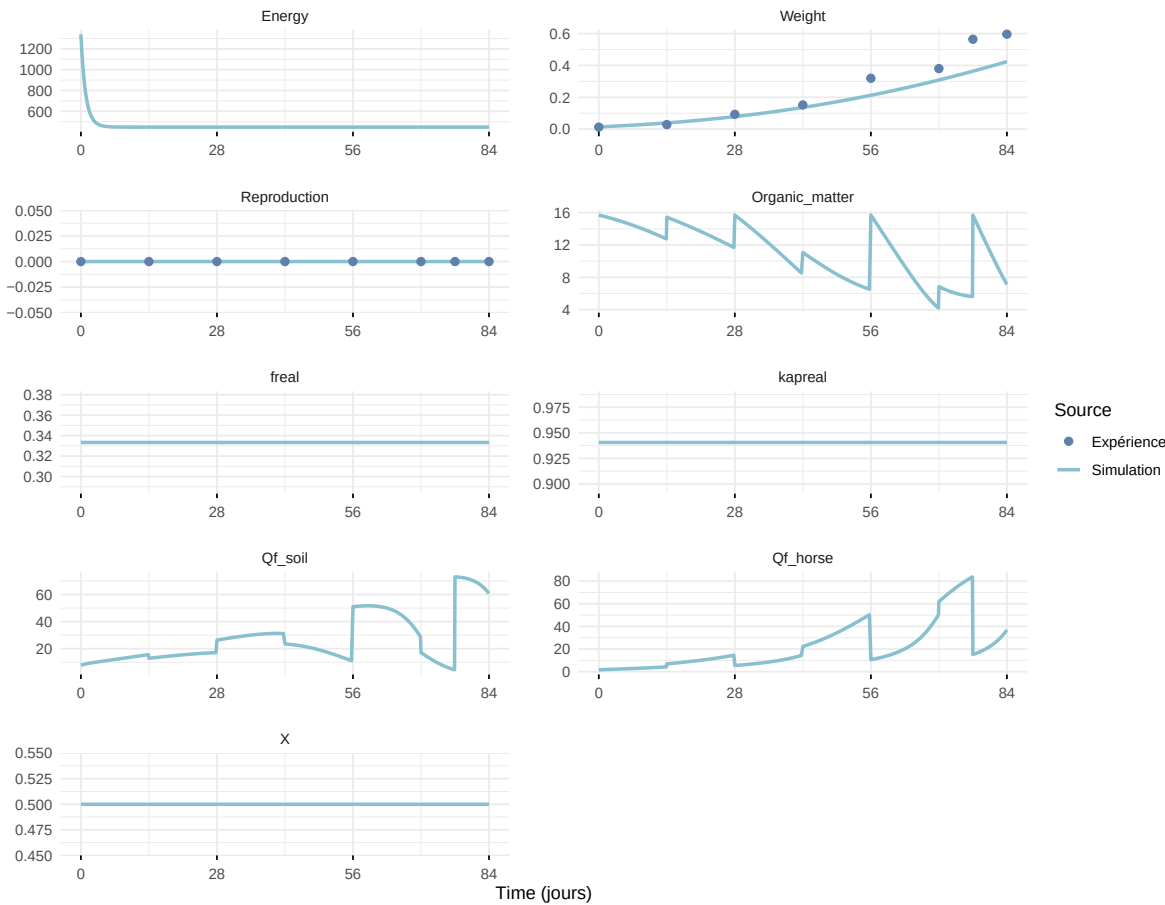


Figure 7: Simulations Weight and Reproduction.

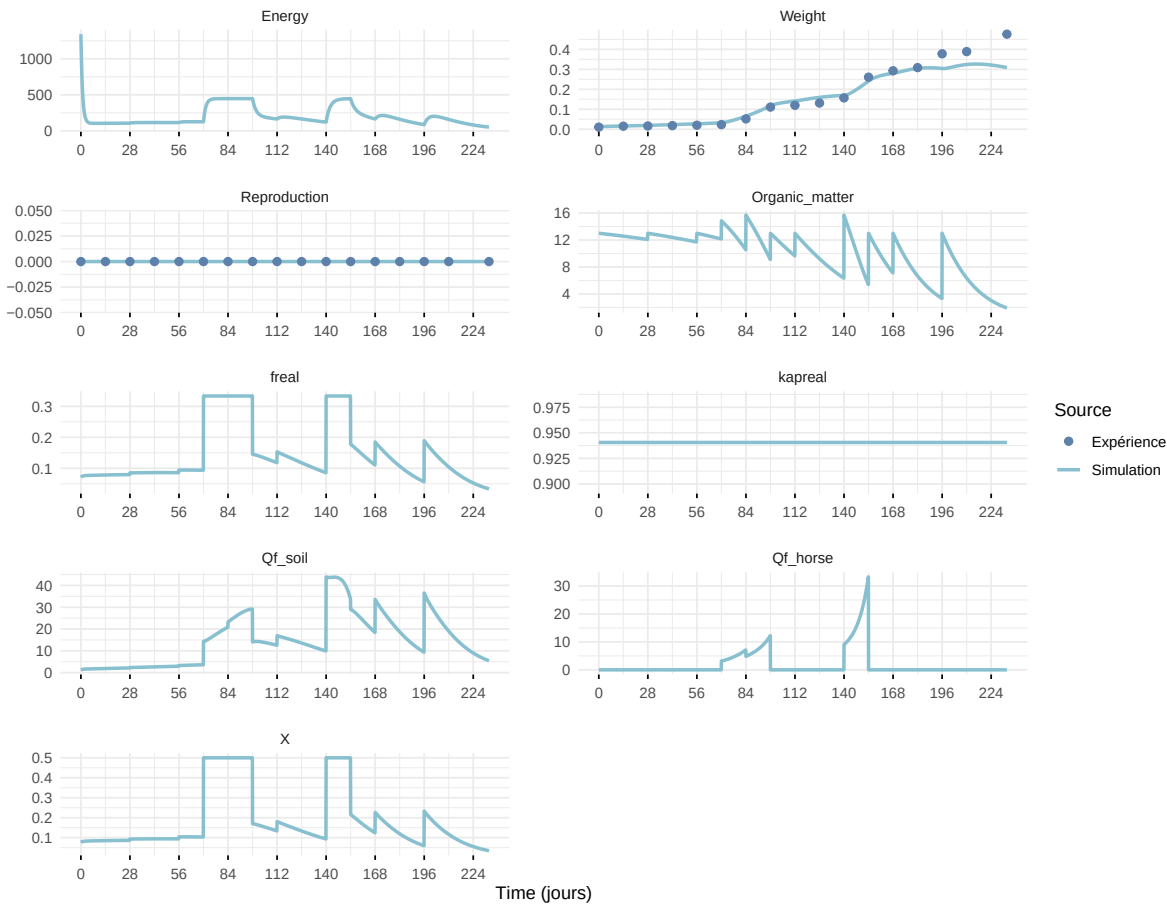
Expérience : Bart2019_XP1_2



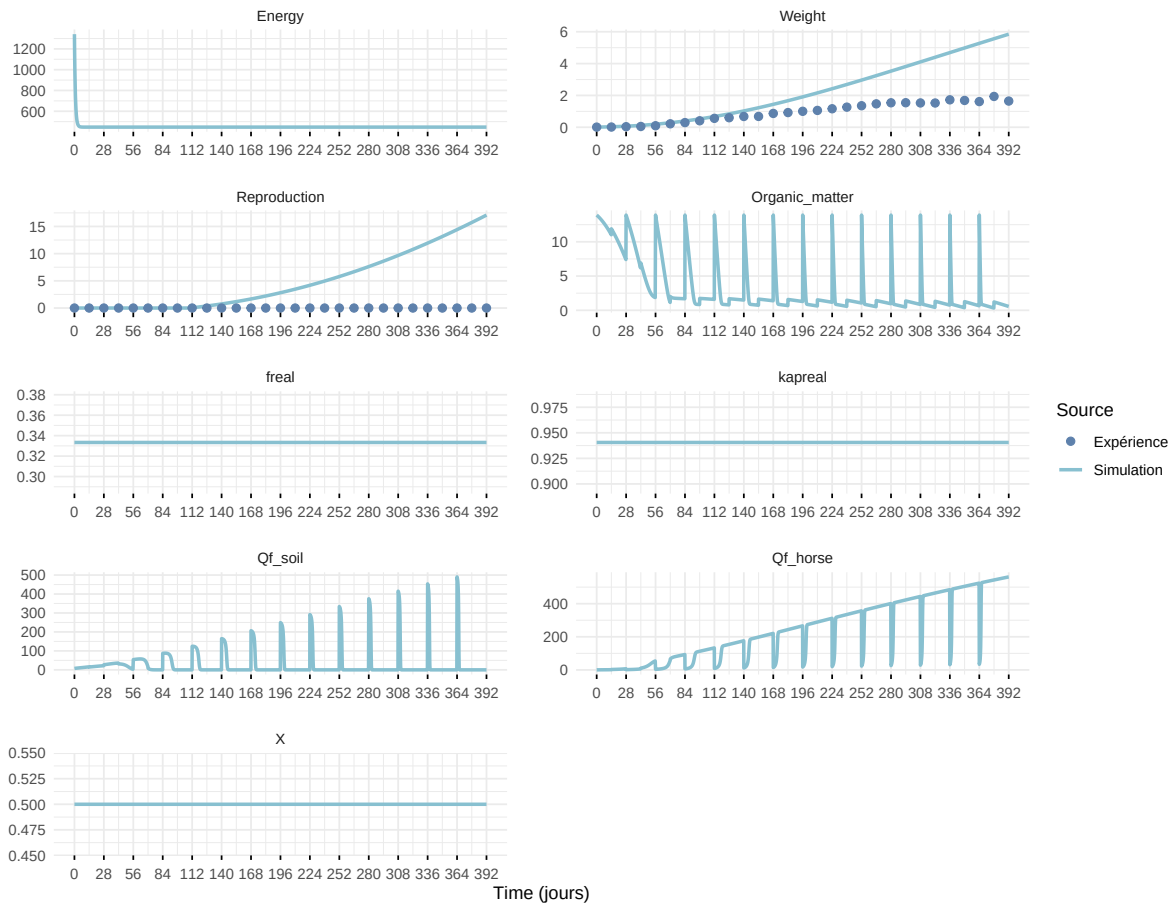
Expérience : Bart2019_XP2



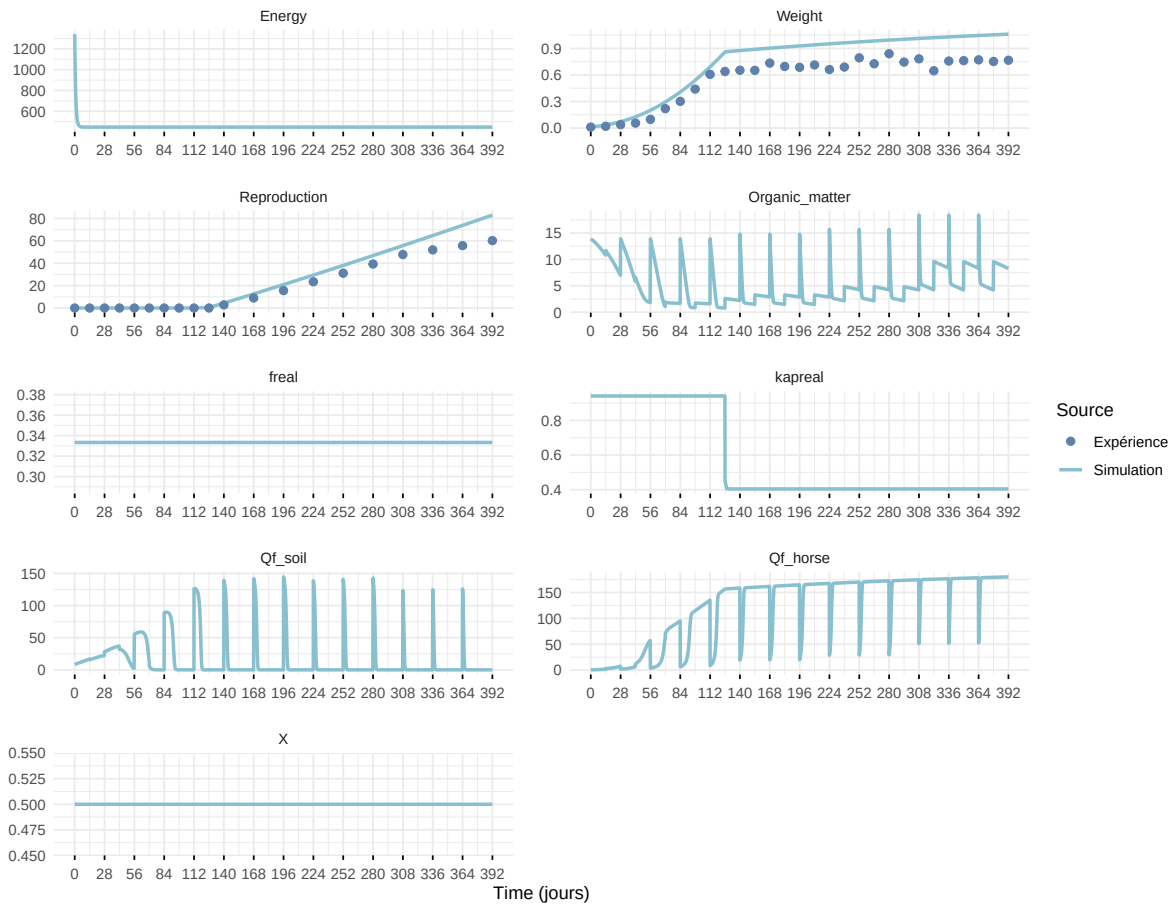
Expérience : Bart2019_XP3



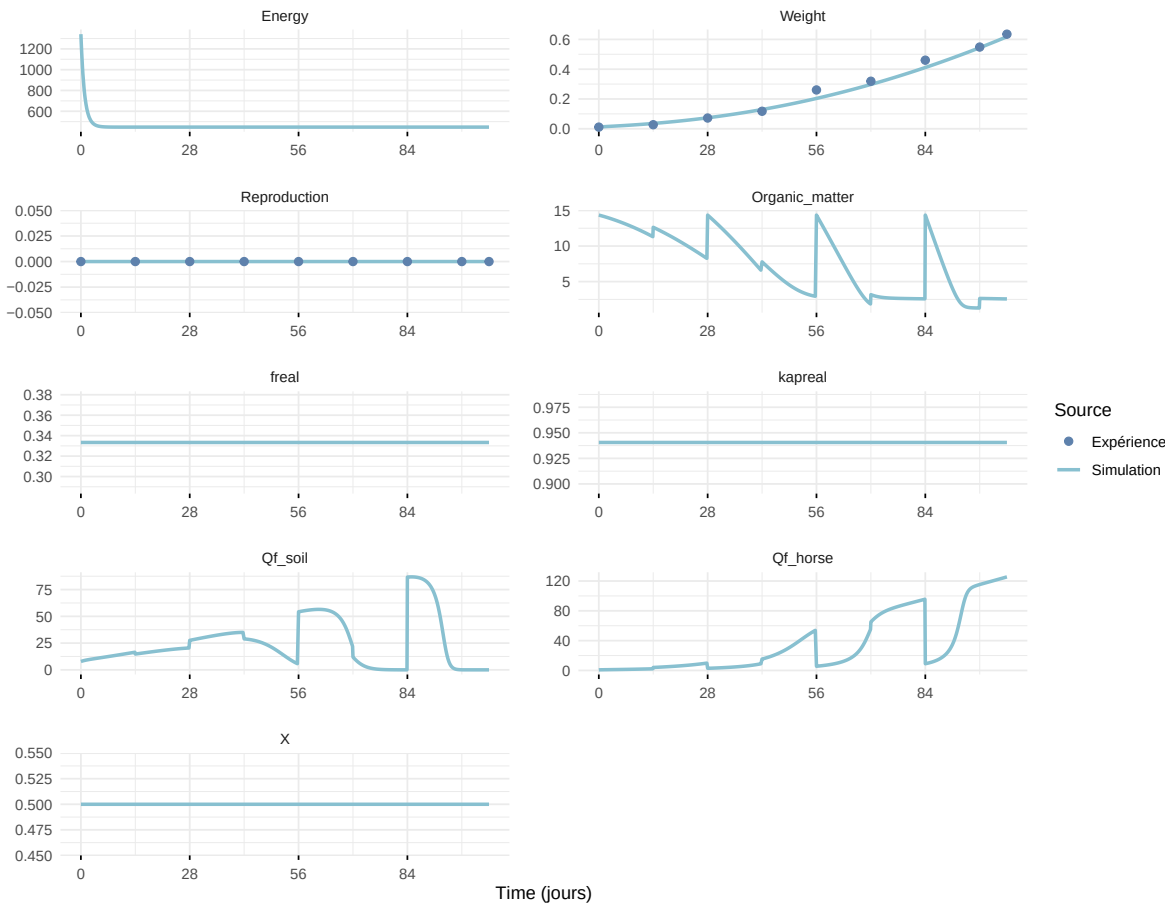
Expérience : Bart2019_XP4_1



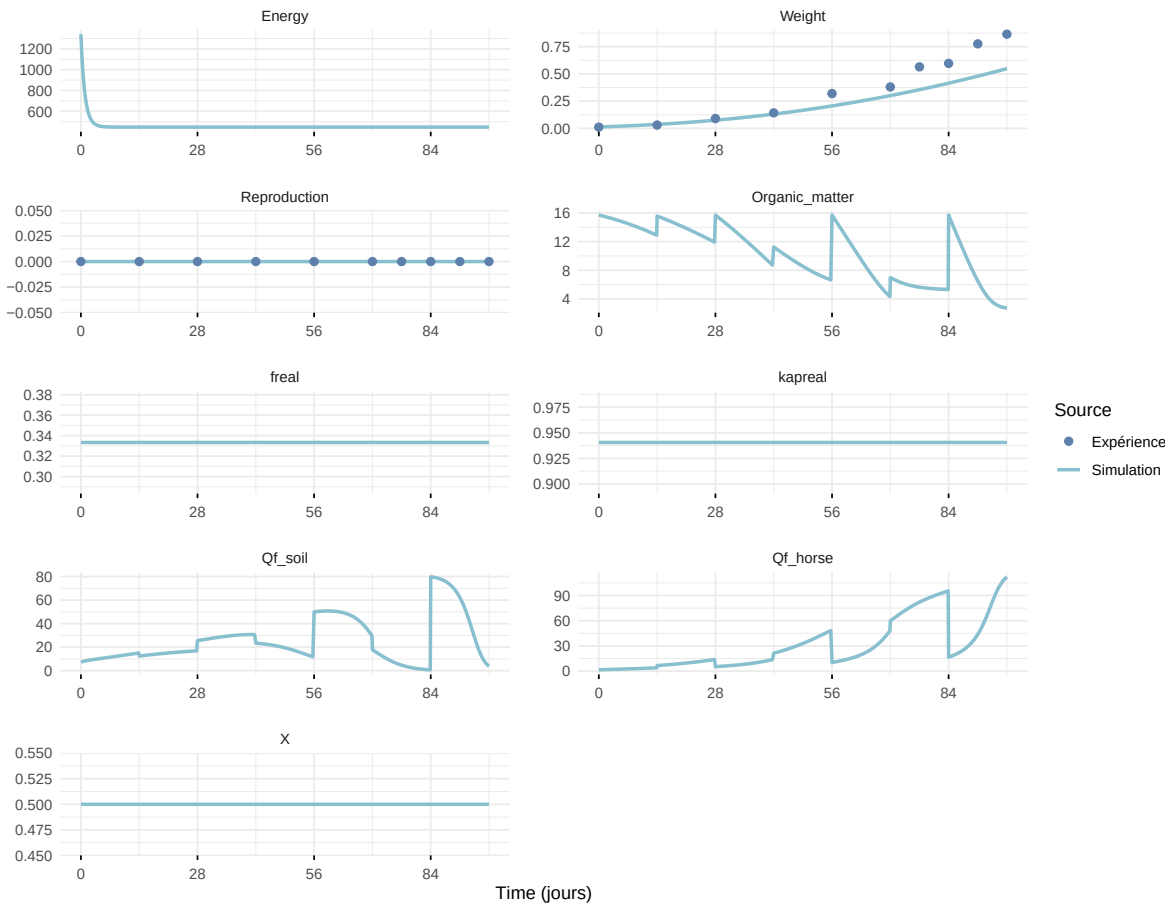
Expérience : Bart2019_XP4_2



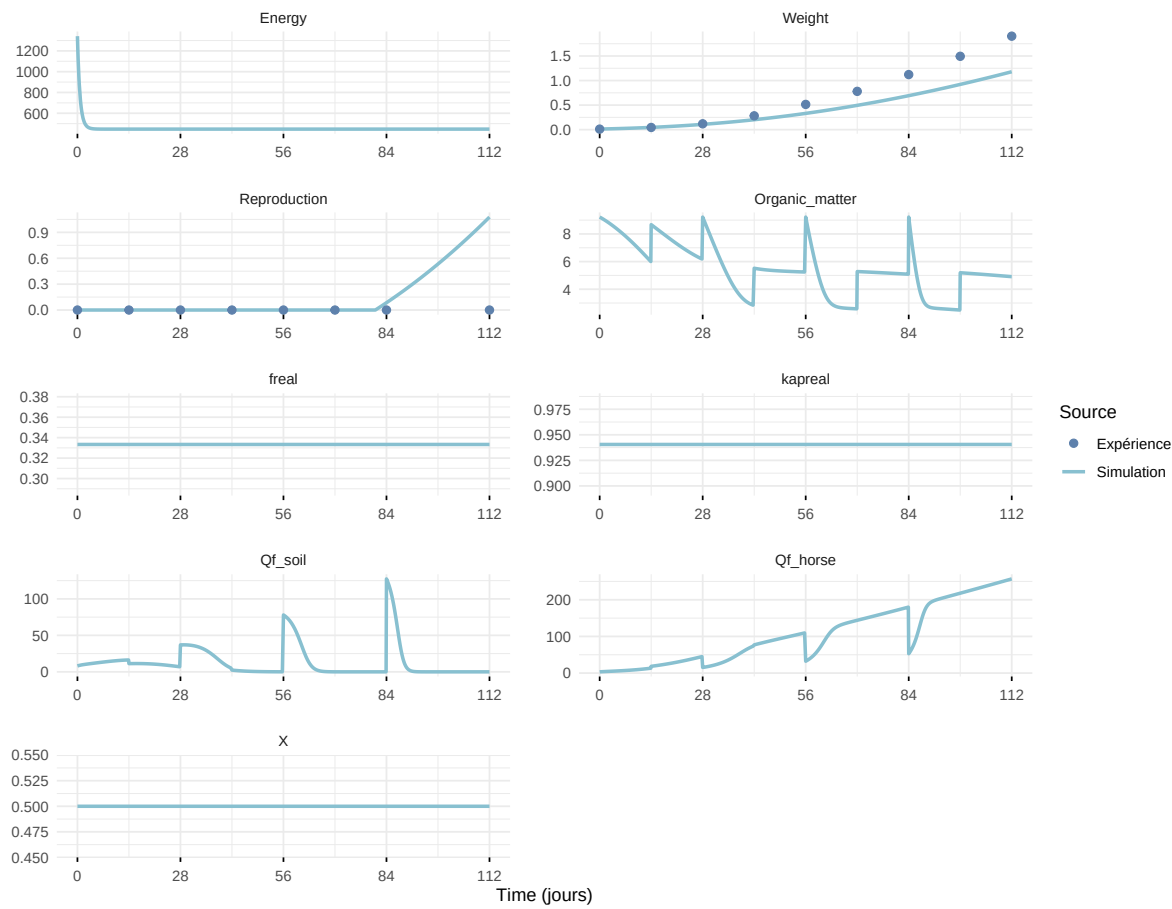
Expérience : Bart2019_XP5



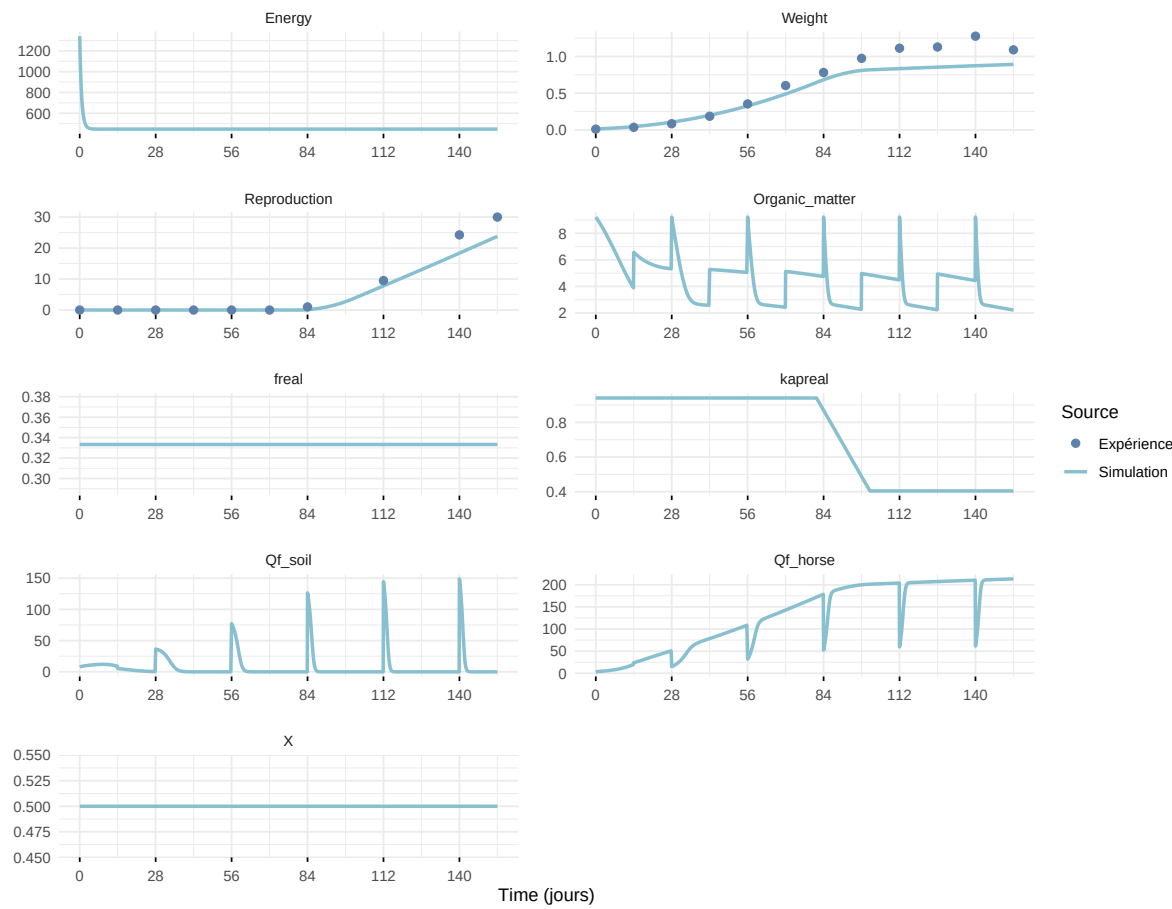
Expérience : Bart2020



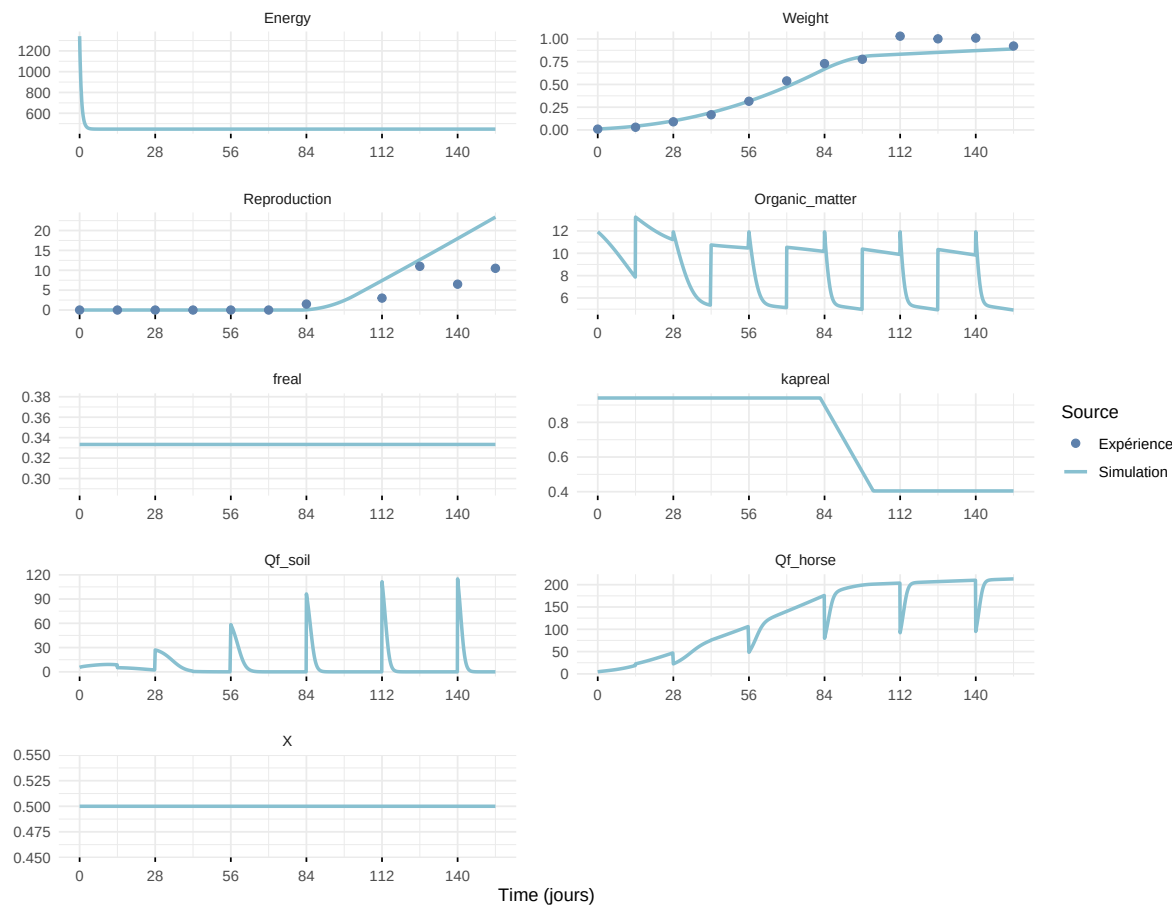
Expérience : Gollot2026_dens_D1



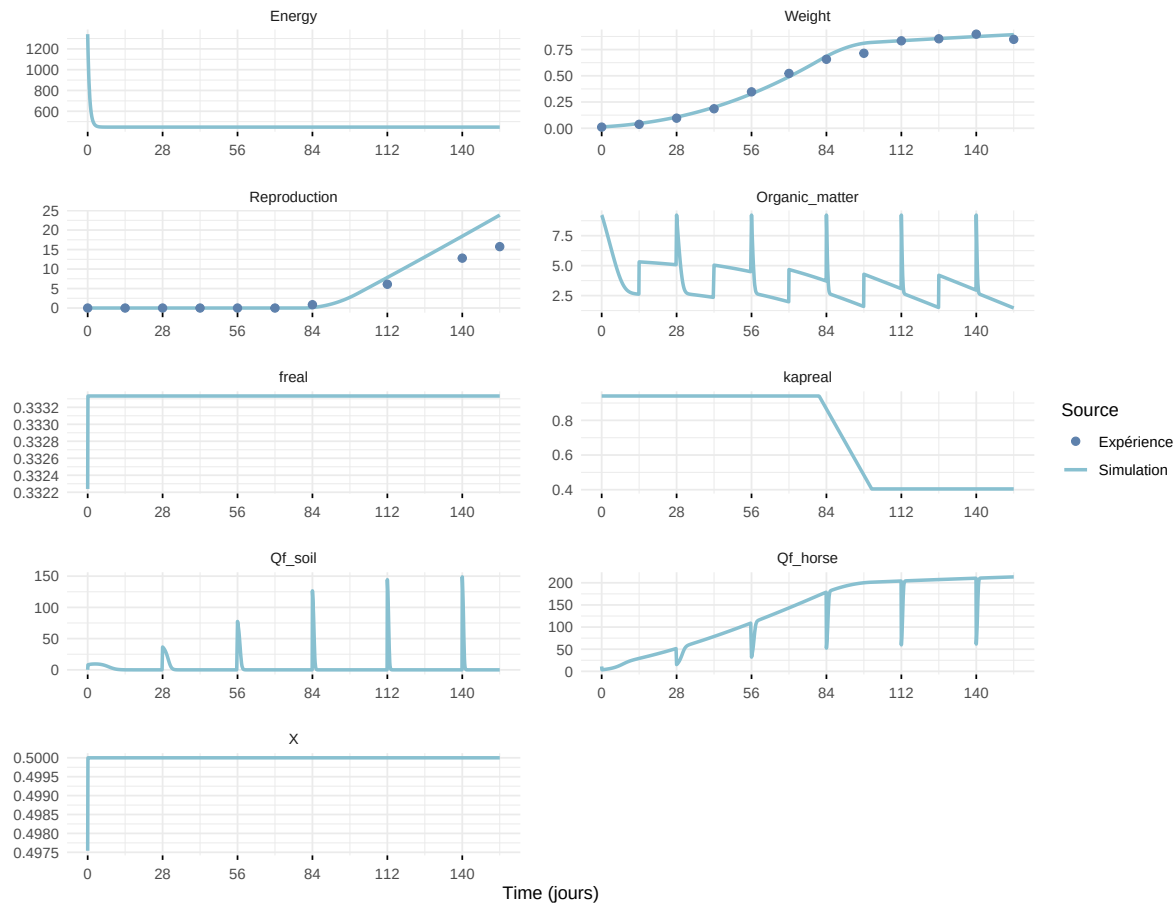
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

